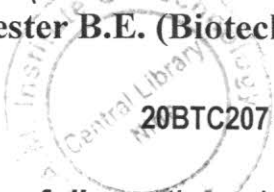


NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations
May - June 2023



20BTC207 - INDUSTRIAL BIOTECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Elaborate on the steps involved in the production of wine.	10	L*2	1	1
	b)	Comment on microbial biofuel production via ethanol fermentation.	10	L3	1	1
2.	a)	Write briefly on: i) Traditional and modern biotechnology ii) Types of culture media.	10	L2	1	1
	b)	Comment on the commercial importance of amino acids.	10	L3	2	1
3.	a)	Write an explanatory note on the stages of the fermentation process.	10	L1	2	1
	b)	What are the main utilities of a fermenter? Describe the functions of the main components of the fermenter.	10	L2	2	1
		Unit – II				
4.	a)	Discuss steps involved in the production process of streptomycin.	10	L3	3	1
	b)	Write short notes on the following: i) Polyhydroxy alkanoates ii) Macrolides.	10	L2	3	1
5.	a)	With illustration explain the production process of industrial enzymes.	10	L2	3	1
	b)	Give a detailed account on the industrial production of lipase.	10	L3	4	1
6.	a)	Write an explanatory note on the production of nisin by <i>Lactococcus lactis</i> .	10	L1	4	1
	b)	Write in detail about the production process of cheese.	10	L3	4	1
		Unit – III				
7.	a)	Write an explanatory note on the production of vaccines.	10	L2	5	1
	b)	How biotechnological tool will help in enhancing the quality of natural silk?	10	L3	5	1
8.	a)	Discuss human growth hormone in detail.	10	L2	5	1
	b)	Elaborate on the steps involved in the production process of human insulin.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
